WEEK 4 OF 15 · STAGE 1: FOUNDATIONS

Economics primer

How prices decide what gets built

New R skill Writing functions; if and for; solving one equation numerically

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	**Week 4** **Economics primer**	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Explain how a market price balances supply and demand
2. Add a carbon price to a technology's cost and say which option wins
3. Discount a future cost to today's value and explain why the rate matters
4. Write a small R function, a loop, and solve one equation with uniroot

? **Big question** If solar is clean but coal is cheaper, which one does the economy choose, and what can change that choice?

Coal, gas or solar?

A city needs electricity. Coal costs 60, gas 70, solar 80 dollars per MWh. Which gets built?



What would have to change for solar to win?

AN EVERYDAY PICTURE

A market is like an auction that never stops



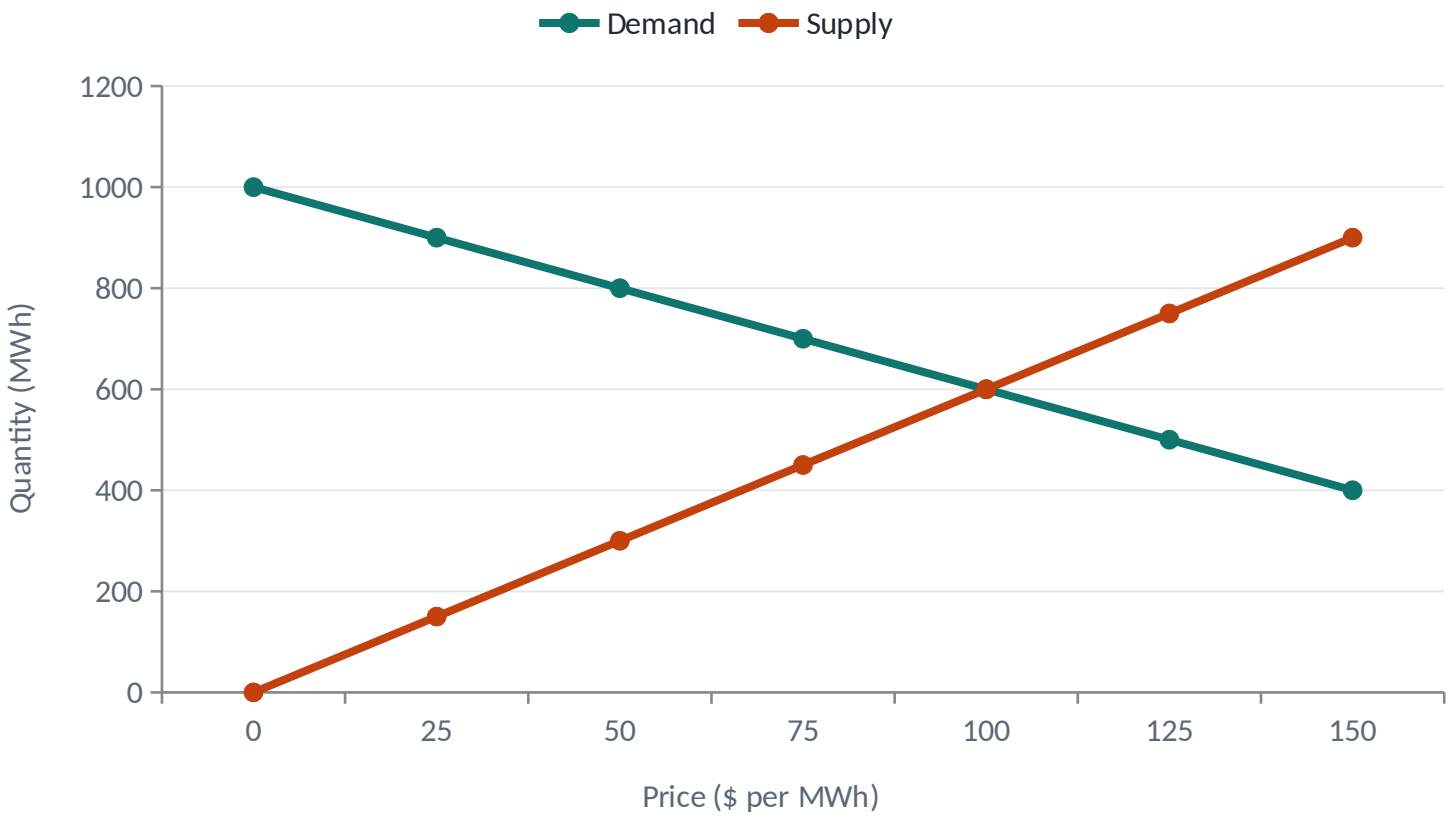
Think of a farmers' market at closing time. If tomatoes pile up unsold, sellers cut the price. If they run out by noon, they raise it next week. The price keeps moving until what is brought equals what is bought.

What it means for models

1. Buyers want less when the price is high
2. Sellers offer more when the price is high
3. The price where both agree is the equilibrium

PICTURE THE NUMBERS

Supply and demand cross at one price



Illustrative numbers from the week 4 lab

Read it like this

1. Demand: 1000 minus 4 × price
2. Supply: 6 × price
3. They meet at a price of 100 and a quantity of 600

Worked example: finding the equilibrium by hand

Problem Demand = 1000 − 4p. Supply = 6p. Find the price p where they are equal.

1. Set them equal: 1000 − 4p = 6p
2. Move the p terms to one side: 1000 = 10p
3. Divide by 10: p = 100
4. Quantity: 6 × 100 = 600

Answer

Price 100 $/MWh, quantity 600 MWh

HOW IT WORKS

A carbon price adds cost in proportion to emissions

1

Start cost

What the plant costs to run, in $/MWh

2

Emissions

Tonnes of CO2 per MWh (the emission factor)

3

Carbon charge

Emission factor × carbon price

4

New cost

Start cost + carbon charge

New cost = base + ef × tax

SIDE BY SIDE

With a $50 carbon price the winner changes

Technology	Cost $/MWh	tCO2 per MWh	With $50/tCO2
Coal	60	0.9	60 + 0.9 × 50 = 105
Gas	70	0.4	70 + 0.4 × 50 = 90
Solar	80	0	80 + 0 × 50 = 80

Cheapest without a carbon price: coal. Cheapest with it: solar. Illustrative numbers.

Quick check

At a carbon price of $22 per tonne, which technology is cheapest?

A Coal (79.8)

B Gas (78.8)

C Solar (80)

D They all cost the same

Quick check: answer

At a carbon price of $22 per tonne, which technology is cheapest?

A Coal (79.8)

B **Gas (78.8)** ✓

C Solar (80)

D They all cost the same

COMPARE

Who pays the carbon tax?

Before the tax

- Price: 100 $/MWh
- Quantity sold: 600 MWh
- No carbon charge

With $50/tCO2 (coal supply)

- Price rises to about 127
- Quantity falls to about 492
- Tax per MWh is 45, but price rises only 27, so sellers absorb 18

AN EVERYDAY PICTURE

Discounting: a dollar today is worth more than a dollar later



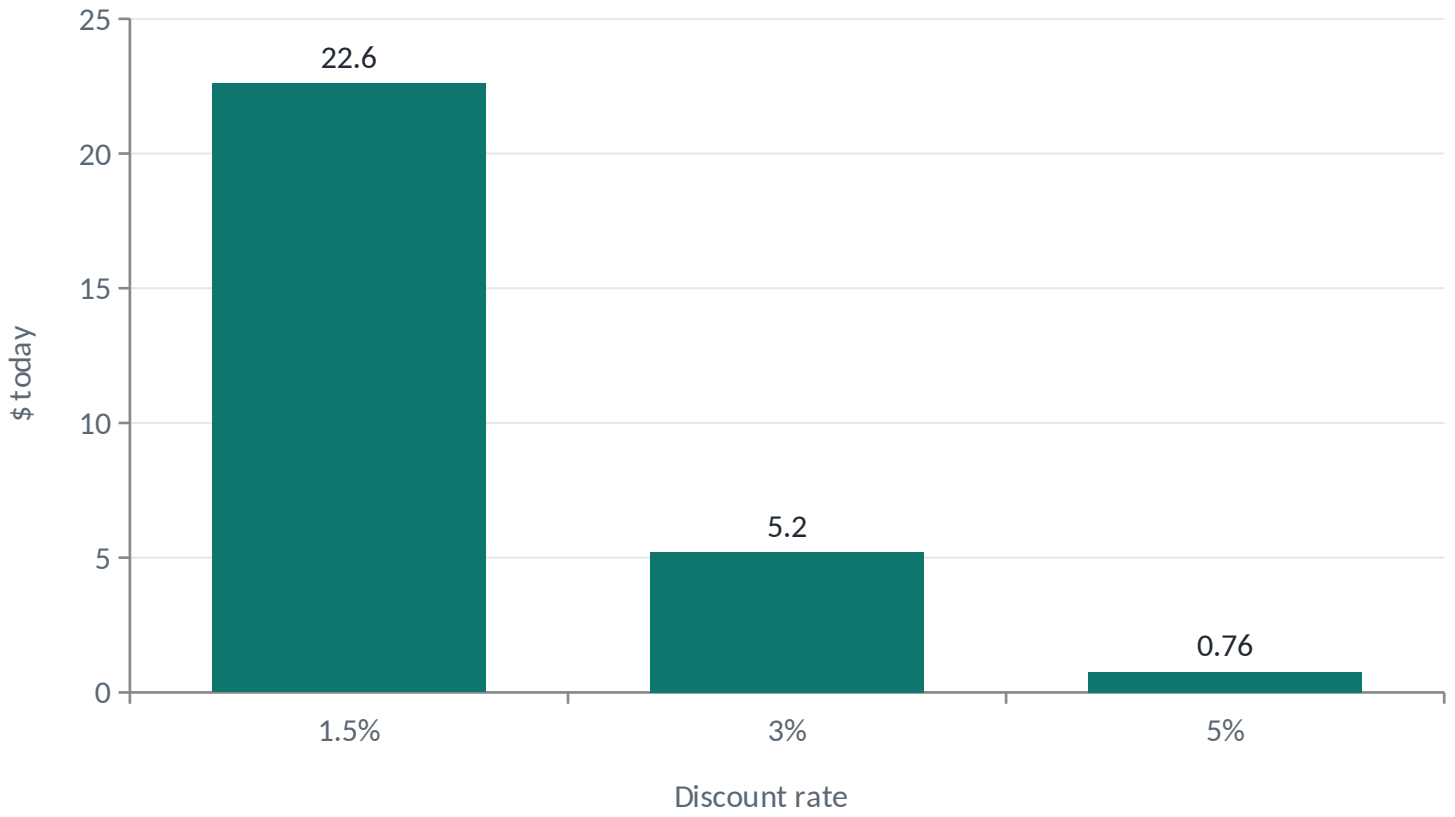
Would you rather get 100 dollars today or 100 dollars in ten years? Almost everyone says today: you could save it and earn interest, or use it now. Discounting turns that feeling into a number.

What it means for models

1. Present value = future amount ÷ (1 + r)^years
2. r is the discount rate
3. The further away, the smaller today's value

PICTURE THE NUMBERS

The discount rate decides how much 2120 counts today



Same future damage, thirty times smaller at 5% than at 1.5%

Read it like this

1. At 1.5%: 100 ÷ 1.015^100 ≈ 22.6
2. At 3%: 100 ÷ 1.03^100 ≈ 5.2
3. At 5%: 100 ÷ 1.05^100 ≈ 0.76

WATCH OUT

Common mix-ups

Not quite	Better
A carbon price makes every technology more expensive	Only options that emit. Solar's cost is unchanged because its emission factor is zero.
The cheapest technology takes 100% of the market	In this toy, yes. Real models like GCAM share the market by relative cost (week 7).
A high discount rate means we care more about the future	The opposite: a high rate shrinks future values, so the future counts less.

Words to add to the class glossary

Equilibrium The price at which the amount people want to buy equals the amount producers want to sell.

Carbon price A charge per tonne of CO2 emitted. It makes high-emitting options more expensive in proportion to their emissions.

Emission factor Tonnes of CO2 released per unit of output, for example 0.9 tCO2 per MWh from a coal plant.

Discount rate The yearly rate used to convert future money into today's value. A higher rate makes the future count less.

Present value What a future amount is worth today: the future amount divided by (1 + r) to the power of the number of years.

Lab 4: a toy energy market in R

Write a cost function, loop over carbon prices to find the cheapest technology, then find the market price with and without a tax.

Script: labs/lab4.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

LAB · STEP 1 OF 4

Write a cost function

A function has a name, inputs in brackets and a body in braces. The last line is what it gives back.

```
cost <- function(base, ef, tax) {
  if (tax < 0) stop("tax is negative")
  base + ef * tax
}
cost(60, 0.9, 50)   # coal at a tax of 50: expect 105
```

Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Loop over four carbon prices

A for loop runs its body once for each value. which.min picks the cheapest option.

```
for (tax in c(0, 22, 30, 50)) {
  costs <- c(coal  = cost(60, 0.9, tax),
             gas   = cost(70, 0.4, tax),
             solar = cost(80, 0, tax))
  print(names(which.min(costs)))
}
# expect: coal, gas, solar, solar
```



Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Describe the market

Demand falls as price rises; supply rises with price but the tax is taken off what sellers keep. gap is demand minus supply.

```
demand <- function(p) 1000 - 4 * p
supply <- function(p, tax) 6 * (p - 0.9 * tax)
gap    <- function(p, tax) demand(p) - supply(p, tax)
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

LAB · STEP 4 OF 4

Let R find the equilibrium

uniroot searches between 0 and 500 for the price where gap is zero, the same search a market model does.

```
p0  <- uniroot(gap, c(0, 500), tax = 0)$root   # ~100
p50 <- uniroot(gap, c(0, 500), tax = 50)$root  # ~127
demand(c(p0, p50))                             # ~600, ~492
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Did it work?

You should see

- The loop prints coal, gas, solar, solar
- p0 is about 100 and p50 about 127
- Quantity falls from 600 to about 492

Worksheet questions

1. Of the 45 $/MWh tax, how much do buyers pay and how much do sellers absorb?
2. At what carbon price does solar beat gas? Show the algebra.
3. Change demand to 1000 − 2p. What happens to the price rise, and who now carries more of the tax?

Take-home messages

1. Prices move until supply equals demand; that crossing is the equilibrium
2. A carbon price adds emission factor × price to each option's cost, so clean options gain
3. Gas beats coal above $20/t and solar beats gas above $25/t in our toy numbers
4. Discounting divides future values by (1 + r)^years, so the rate decides how much the future counts

Next: week 5

Week 5 joins the climate chain, the economy and discounting into one small model: DICE.

Practise

Interactive course, week 4: try the market explorer and the 5-question quiz.

Readings: Instructor's notes on markets, discounting and carbon pricing; R for Data Science: functions and iteration chapters